

AI BIAS AUDITOR v3.0

USER GUIDE - COMPREHENSIVE

Panduan Lengkap Penggunaan, Interpretasi, dan Metodologi

Audit Teknologi dan Sistem Informasi

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1. PENGENALAN AI BIAS AUDITOR

1.1. Latar Belakang

AI Bias Auditor v3.0 adalah tool profesional untuk mendeteksi dan mengukur bias sistemik dalam sistem credit scoring berbasis AI. Tool ini dirancang untuk membantu auditor, data scientist, compliance officer, dan stakeholder terkait dalam:

- Mengidentifikasi diskriminasi berdasarkan umur (age discrimination)
- Mengukur disparate impact dengan metode statistik yang robust
- Memvisualisasikan pola bias dengan 18+ interactive charts
- Memberikan rekomendasi actionable untuk fairness intervention
- Memastikan compliance dengan regulasi (EEOC, OJK, ISO 24027)

1.2. Fitur Utama

- 18+ Interactive Visualizations (Heatmap, Violin Plot, Sunburst, Radar Chart)
- 8 Statistical Tests (T-Test, Mann-Whitney U, Cohen's d, DI Ratio, dll)
- 6 ML Fairness Metrics (DPD, EOD, Predictive Parity)
- Real-time Progress Tracking
- Multiple Export Formats (CSV, TXT, JSON)
- Professional Academic Documentation

2. GLOSSARY - DEFINISI ISTILAH LENGKAP

Applicant

Individu yang mengajukan permohonan kredit/pinjaman kepada lembaga keuangan. Dalam dataset, setiap applicant memiliki ID unik dan atribut seperti Age, Income, Credit Score.

Approval Rate

Persentase applicants yang disetujui (loan approved) dari total applicants dalam suatu kelompok. Formula: $(\text{Jumlah Approved} / \text{Total Applicants}) \times 100\%$. Contoh: Jika 72 dari 100 applicants disetujui, approval rate = 72%. Metric ini digunakan untuk membandingkan treatment antara kelompok Young dan Old.

Credit Score

Nilai numerik (skala 300-850) yang merepresentasikan kelayakan kredit seseorang berdasarkan berbagai faktor seperti payment history, income, debt-to-income ratio, credit history length, dll. Semakin tinggi skor, semakin rendah risiko gagal bayar. Kategori standard: 300-500 (High Risk), 500-600 (Medium Risk), 600-700 (Low Risk), 700-850 (Very Low Risk). Threshold approval biasanya di 600 (applicants dengan score ≥ 600 disetujui).

Age Group

Klasifikasi applicants berdasarkan umur untuk analisis bias: Young (umur ≤ 45 tahun) dan Old (umur > 45 tahun). Threshold 45 tahun dipilih berdasarkan Age Discrimination in Employment Act (ADEA) yang melindungi workers berumur 40+ tahun. Dalam credit scoring, threshold ini digunakan untuk mendeteksi age discrimination (ageism).

Bias (Bias Sistemik)

Perlakuan tidak adil secara sistematis terhadap kelompok tertentu (protected group) dalam proses pengambilan keputusan AI. Bias dapat terjadi karena: (1) Data bias (historical discrimination), (2) Algorithm bias (flawed model design), (3) Proxy discrimination (fitur yang berkorelasi dengan protected attribute). Dalam konteks ini: diskriminasi berdasarkan umur (age discrimination / ageism).

Disparate Impact (DI) Ratio

Rasio selection rate (approval rate) kelompok minoritas/protected dibanding kelompok mayoritas. Formula: (Selection Rate Minoritas / Selection Rate Mayoritas). Dalam konteks ini: (Approval Rate Old / Approval Rate Young). Standard legal: DI Ratio ≥ 0.80 (80% Rule dari EEOC Uniform Guidelines 1978). Jika DI < 0.80 , indikasi adverse impact yang dapat digugat secara hukum. Contoh: Jika Young approval = 72% dan Old approval = 29%, DI = $29/72 = 0.40$ (FAIL!).

p-value

Probabilitas bahwa perbedaan yang diamati terjadi karena kebetulan (random chance) jika null hypothesis benar. Null hypothesis: Tidak ada perbedaan antara kelompok Young dan Old. Interpretasi: $p < 0.05$ = perbedaan signifikan secara statistik (reject null hypothesis, bukan kebetulan). $p < 0.01$ = sangat signifikan. $p < 0.001$ = extremely significant. $p \geq 0.05$ = tidak ada perbedaan yang signifikan (fail to reject null hypothesis). Threshold standard: alpha = 0.05 (95% confidence level).

T-Test (Independent Samples T-Test)

Parametric statistical test untuk membandingkan mean (rata-rata) dari 2 independent groups. Asumsi: (1) Data berdistribusi normal, (2) Equal/unequal variance. Menghasilkan: t-statistic dan p-value. Digunakan untuk menguji apakah perbedaan rata-rata credit score antara Young dan Old signifikan.

Mann-Whitney U Test

Non-parametric statistical test (distribution-free) untuk membandingkan 2 independent groups. Alternatif T-Test yang tidak mengasumsikan distribusi normal. Lebih robust terhadap outliers dan skewed distributions. Menghasilkan: U-statistic dan p-value. Interpretasi p-value sama dengan T-Test.

Cohen's d (Effect Size)

Ukuran standardized untuk seberapa BESAR perbedaan antara 2 kelompok, independent dari sample size. Formula: $(\text{Mean1} - \text{Mean2}) / \text{Pooled Standard Deviation}$. Interpretasi (Cohen, 1988): $d = 0.2$ (small), $d = 0.5$ (medium), $d = 0.8$ (large), $d > 1.0$ (very large). Contoh: $d = 3.142$ artinya perbedaan 3.14 standard deviations, extremely large effect.

Demographic Parity Difference (DPD)

Selisih selection rate antara 2 kelompok. Formula: $\text{Selection Rate Group A} - \text{Selection Rate Group B}$. Idealnya $\text{DPD} = 0$ (perfect parity). $\text{DPD} > 20\%$ indikasi bias signifikan. Merupakan ML fairness metric untuk demographic parity / statistical parity.

Equal Opportunity Difference (EOD)

Selisih True Positive Rate (TPR) antara 2 kelompok. $\text{TPR} = \text{Sensitivity} = \text{Recall}$. Formula: $\text{TPR Group A} - \text{TPR Group B}$. Fokus pada fairness untuk "qualified individuals" (yang seharusnya approved). Idealnya $\text{EOD} = 0$. Metric ini lebih strict dari Demographic Parity.

Predictive Parity

Rasio Positive Predictive Value (PPV) antara 2 kelompok. $\text{PPV} = \text{Precision}$. Formula: $\text{PPV Group A} / \text{PPV Group B}$. Fokus pada fairness untuk "predicted positives". Idealnya ratio = 1.0 (equal precision across groups).

Proxy Discrimination

Diskriminasi tidak langsung melalui fitur yang berkorelasi tinggi dengan protected attribute. Contoh: `Employment_Years` berkorelasi dengan `Age` (orang tua = lama kerja). Jika model memberikan bobot tinggi pada `Employment_Years`, efektifnya sama dengan diskriminasi `Age`. Deteksi: Correlation Heatmap untuk identifikasi proxy features.

3. CARA BACA METRICS LENGKAP

3.1. Executive Summary Metrics

Executive Summary menampilkan 5 Key Performance Indicators (KPI) di bagian atas dashboard:

1. Total Applicants: Jumlah total applicants dalam dataset.
 - Interpretasi: Semakin besar sample, semakin reliable hasil statistik.
 - Minimum recommended: 100 applicants (30 per group).
2. Young (≤ 45): Jumlah applicants muda + approval rate mereka.
 - Delta indicator: Persentase yang approved.
 - Contoh: 580 applicants, 72.4% approved.
3. Old (> 45): Jumlah applicants tua + approval rate mereka.
 - Delta indicator: Persentase yang approved.
 - Contoh: 420 applicants, 28.6% approved.
4. Bias Status: Indikator utama apakah bias terdeteksi.
 - [!] DETECTED: Ada bias signifikan (action required).
 - [OK] NOT DETECTED: Tidak ada bias signifikan (monitoring).
 - Delta: Approval gap percentage.
5. 80% Rule: Compliance dengan Disparate Impact standard.
 - PASS: DI Ratio ≥ 0.80 (compliant).
 - FAIL: DI Ratio < 0.80 (non-compliant, legal risk).
 - Value: Actual DI Ratio.

3.2. Score Gap (Selisih Skor) - Detail

Score Gap mengukur perbedaan rata-rata credit score antara kelompok Young dan Old.

Formula:

Score Gap = Average Credit Score (Young) - Average Credit Score (Old)

Interpretasi Berdasarkan Magnitude:

- Gap < 25 points: Perbedaan trivial, likely random variation
- Gap 25-50 points: Perbedaan kecil, mungkin wajar tergantung konteks
- Gap 50-100 points: Perbedaan moderat, perlu investigasi lebih lanjut
- Gap 100-150 points: Perbedaan besar, strong indicator of bias
- Gap > 150 points: Perbedaan sangat besar, clear evidence of systematic discrimination

Contoh Perhitungan:

Average Score Young = 632 points

Average Score Old = 439 points

Score Gap = 632 - 439 = 193 points

Interpretasi: Gap 193 points adalah BESAR (>150), clear evidence of age bias.

Catatan:

- Gap positif: Young lebih tinggi (discrimination against Old)
- Gap negatif: Old lebih tinggi (discrimination against Young, rare case)
- Threshold 100 points dipilih berdasarkan praktik industri dan fairness research.

3.3. Approval Gap (Selisih Approval Rate) - Detail

Approval Gap mengukur perbedaan approval rate antara kelompok Young dan Old.

Formula:

$$\text{Approval Gap} = \text{Approval Rate (Young)} - \text{Approval Rate (Old)}$$

Interpretasi Berdasarkan Magnitude:

- Gap < 5%: Perbedaan trivial, acceptable variation
- Gap 5-10%: Perbedaan kecil, relatif fair, monitoring recommended
- Gap 10-20%: Perbedaan moderat, perlu investigation dan possible intervention
- Gap 20-30%: Perbedaan besar, strong evidence of disparate treatment
- Gap > 30%: Perbedaan sangat besar, severe discrimination, immediate action required

Contoh Perhitungan:

Approval Rate Young = 72.4%

Approval Rate Old = 28.6%

$$\text{Approval Gap} = 72.4\% - 28.6\% = 43.8\%$$

Interpretasi: Gap 43.8% adalah SANGAT BESAR (>30%), severe age discrimination.

Hubungan dengan Disparate Impact:

Approval Gap dan DI Ratio saling terkait:

- Jika Approval Gap besar, DI Ratio akan kecil (< 0.80)
- Jika Approval Gap kecil, DI Ratio akan mendekati 1.0

Contoh: Gap 43.8% -> $DI = 28.6/72.4 = 0.395$ (FAIL)

Threshold 20% dipilih berdasarkan fairness literature dan praktik audit.

3.4. T-Test dan p-value - Detail

T-Test menguji null hypothesis: "Tidak ada perbedaan mean antara Young dan Old".

Output T-Test:

1. t-statistic: Ukuran seberapa jauh sample mean dari null hypothesis.
 - t besar (>2 atau <-2): Strong evidence against null hypothesis
 - t kecil (-2 to 2): Weak evidence
2. p-value: Probabilitas mendapatkan hasil ini jika null hypothesis benar.
 - $p < 0.001$: Extremely significant (***) - 99.9% confidence
 - $p < 0.01$: Highly significant (**) - 99% confidence
 - $p < 0.05$: Significant (*) - 95% confidence - THRESHOLD STANDARD
 - $p \geq 0.05$: Not significant (ns) - fail to reject null hypothesis

Interpretasi p-value:

$p = 0.00000001$ artinya:

- Probabilitas hasil ini terjadi secara kebetulan: 0.000001%
- Confidence bahwa perbedaan REAL: 99.999999%
- Conclusion: REJECT null hypothesis, perbedaan SIGNIFIKAN

Catatan Penting:

- p-value menunjukkan SIGNIFIKANSI, bukan MAGNITUDE
- p-value kecil + small effect size = signifikan tapi tidak praktis penting
- p-value kecil + large effect size = signifikan DAN praktis penting
- Selalu kombinasikan p-value dengan effect size (Cohen's d)

Asumsi T-Test:

1. Independent samples (Young dan Old tidak overlap)
2. Normal distribution (dapat direlaksasi dengan sample besar, $n > 30$)
3. Homogeneity of variance (dapat gunakan Welch's t-test jika tidak terpenuhi)

3.5. Mann-Whitney U Test - Detail

Mann-Whitney U Test adalah alternatif non-parametric untuk T-Test.

Kapan Digunakan:

- Data tidak berdistribusi normal (skewed, heavy-tailed)
- Ada outliers yang ekstrem
- Sample size kecil ($n < 30$)
- Data ordinal (bukan interval/ratio)

Cara Kerja:

Test ini membandingkan RANK (peringkat) data, bukan nilai actual.

- Rank semua data dari terendah ke tertinggi
- Bandingkan sum of ranks antara 2 kelompok
- Hitung U-statistic

Output:

1. U-statistic: Ukuran perbedaan rank sum
2. p-value: Interpretasi sama dengan T-Test

Keuntungan:

- Lebih robust terhadap outliers
- Tidak asumsi distribusi
- Lebih conservative (less likely false positive)

Kekurangan:

- Less powerful dibanding T-Test (jika data memang normal)
- Lebih sulit interpretasi (rank-based)

Rekomendasi:

Gunakan KEDUANYA (T-Test DAN Mann-Whitney U):

- Jika KEDUANYA signifikan: Strong evidence
- Jika hanya salah satu signifikan: Investigate further
- Jika KEDUANYA tidak signifikan: No evidence of bias

3.6. Cohen's d (Effect Size) - Detail

Cohen's d mengukur MAGNITUDE of difference antara 2 kelompok.

Formula:

$d = (\text{Mean1} - \text{Mean2}) / \text{Pooled Standard Deviation}$

$\text{Pooled SD} = \sqrt{((n1-1)*SD1^2 + (n2-1)*SD2^2) / (n1+n2-2)}$

Interpretasi (Cohen, 1988):

d = 0.0 - 0.2: Trivial/Negligible effect

d = 0.2 - 0.5: Small effect

d = 0.5 - 0.8: Medium effect

d = 0.8 - 1.2: Large effect

d > 1.2: Very large effect

d > 2.0: Huge effect (extremely rare in social sciences)

Contoh:

d = 3.142 artinya:

- Perbedaan mean = 3.14 standard deviations
- Extremely large effect (jauh di atas threshold 0.8)
- Praktis signifikan: Perbedaan terlihat jelas dalam praktik

Keuntungan Cohen's d:

- Independent dari sample size (berbeda dengan p-value)
- Ukuran practical significance, bukan statistical significance
- Dapat dibandingkan across studies
- Interpretasi lebih intuitif

Kombinasi p-value dan Cohen's d:

- $p < 0.05 + d < 0.2$: Signifikan tapi tidak praktis penting
- $p < 0.05 + d > 0.8$: Signifikan DAN praktis penting (ACTION REQUIRED)
- $p > 0.05 + d > 0.8$: Tidak signifikan, mungkin sample size kecil
- $p > 0.05 + d < 0.2$: Tidak signifikan dan tidak penting

3.7. Disparate Impact Ratio (80% Rule) - Detail

Disparate Impact Ratio adalah legal standard untuk mendeteksi discrimination.

Sejarah:

EEOC Uniform Guidelines on Employee Selection Procedures (1978) menetapkan "Four-Fifths Rule" atau "80% Rule" sebagai threshold untuk adverse impact dalam employment selection.

Formula:

DI Ratio = (Selection Rate Protected Group / Selection Rate Reference Group)

Dalam konteks ini: DI = (Approval Rate Old / Approval Rate Young)

Legal Standard:

- DI $\geq$ 0.80: PASS (Presumption of no adverse impact)
- DI < 0.80: FAIL (Adverse impact, potential legal violation)
- DI < 0.50: Severe adverse impact (high legal risk)

Contoh Perhitungan:

Approval Rate Young = 72.4%

Approval Rate Old = 28.6%

DI Ratio = $28.6 / 72.4 = 0.395$

Interpretation: $0.395 < 0.80 \rightarrow$ FAIL

Artinya: Old group hanya 39.5% sebesar kemungkinan Young group untuk approved.

Ini di bawah 80% threshold, indikasi adverse impact yang dapat digugat.

Legal Implications:

- DI < 0.80: Prima facie evidence of discrimination (burden of proof shifts to defendant)
- Defendant harus membuktikan: (1) Business necessity, (2) Job-related, (3) No less discriminatory alternative
- Jika gagal membuktikan: Liable for discrimination (fines, damages, corrective action)

Aplikasi di Credit Scoring:

Meskipun 80% Rule berasal dari employment law, prinsip yang sama applicable untuk:

- Fair lending practices
- Consumer protection law
- AI ethics and fairness standards (ISO 24027)
- Corporate governance and ESG (Environmental, Social, Governance)

4. CONTOH INTERPRETASI LENGKAP

4.1. Skenario 1: Bias Terdeteksi (High Severity)

Dataset: 1000 applicants (580 Young, 420 Old)

Metrics:

- Young Approval Rate: 72.4%
- Old Approval Rate: 28.6%
- Approval Gap: 43.8%
- Young Avg Score: 632
- Old Avg Score: 439
- Score Gap: 193 points
- DI Ratio: 0.395
- T-Test p-value: 0.00000001
- Cohen's d: 3.142

Interpretasi:

1. Approval Gap 43.8% >> 20% threshold -> BIAS SIGNIFIKAN
2. Score Gap 193 points >> 100 threshold -> BIAS SIGNIFIKAN
3. DI Ratio 0.395 << 0.80 -> FAIL 80% Rule, LEGAL VIOLATION
4. p-value 0.00000001 << 0.05 -> EXTREMELY SIGNIFICANT
5. Cohen's d 3.142 >> 0.8 -> VERY LARGE EFFECT

Kesimpulan:

SEVERE AGE DISCRIMINATION DETECTED. Semua metrics menunjukkan bias yang jelas dan signifikan. System memberikan treatment yang sangat berbeda terhadap applicants berumur >45 tahun. DI Ratio jauh di bawah 80% threshold, mengindikasikan adverse impact yang melanggar hukum.

Severity Level: HIGH (Immediate action required)

Rekomendasi:

1. Immediate: Suspend penggunaan model untuk approval decisions
2. Investigation: Audit model logic, identifikasi source of bias
3. Legal review: Konsultasi legal team terkait compliance risk
4. Remediation: Retrain model dengan fairness constraints
5. Monitoring: Deploy continuous bias monitoring dashboard
6. Transparency: Disclose bias findings ke stakeholders
7. Compensation: Review past decisions, possible remediation untuk affected applicants

4.2. Skenario 2: Bias Tidak Terdeteksi (Fair System)

Dataset: 1000 applicants (520 Young, 480 Old)

Metrics:

- Young Approval Rate: 65.2%
- Old Approval Rate: 62.8%
- Approval Gap: 2.4%
- Young Avg Score: 615
- Old Avg Score: 608
- Score Gap: 7 points
- DI Ratio: 0.963
- T-Test p-value: 0.234
- Cohen's d: 0.08

Interpretasi:

1. Approval Gap 2.4% << 20% threshold -> NO SIGNIFICANT BIAS
2. Score Gap 7 points << 100 threshold -> NO SIGNIFICANT BIAS
3. DI Ratio 0.963 >> 0.80 -> PASS 80% Rule, COMPLIANT
4. p-value 0.234 >> 0.05 -> NOT SIGNIFICANT
5. Cohen's d 0.08 << 0.8 -> NEGLIGIBLE EFFECT

Kesimpulan:

NO BIAS DETECTED. System menunjukkan treatment yang fair dan equitable terhadap kedua age groups. Perbedaan approval rate dan credit score sangat kecil dan tidak signifikan secara statistik. DI Ratio di atas 80% threshold, mengindikasikan compliance dengan fairness standards.

Severity Level: NONE (System is fair)

Rekomendasi:

1. Continue monitoring: Maintain bias monitoring sebagai best practice
2. Documentation: Document fairness findings untuk audit trail
3. Periodic review: Re-audit setiap quarter atau saat model update
4. Transparency: Communicate fairness metrics ke stakeholders
5. Best practices: Share fairness approach sebagai benchmark

4.3. Skenario 3: Bias Moderat (Medium Severity)

Dataset: 500 applicants (280 Young, 220 Old)

Metrics:

- Young Approval Rate: 68.5%
- Old Approval Rate: 52.3%
- Approval Gap: 16.2%
- Young Avg Score: 625
- Old Avg Score: 585
- Score Gap: 40 points
- DI Ratio: 0.763
- T-Test p-value: 0.012
- Cohen's d: 0.42

Interpretasi:

1. Approval Gap 16.2% mendekati 20% threshold -> MODERAT, perlu monitoring
2. Score Gap 40 points < 100 threshold -> Tidak major concern, tapi notable
3. DI Ratio 0.763 < 0.80 -> FAIL 80% Rule (borderline)
4. p-value 0.012 < 0.05 -> SIGNIFICANT
5. Cohen's d 0.42 (0.2-0.5) -> SMALL to MEDIUM EFFECT

Kesimpulan:

MODERATE BIAS DETECTED. System menunjukkan perbedaan treatment yang signifikan secara statistik, meskipun magnitude-nya tidak sebesar high severity case. DI Ratio sedikit di bawah 80% threshold (76.3%), mengindikasikan potential adverse impact. Effect size (Cohen's d = 0.42) termasuk small-to-medium.

Severity Level: MEDIUM (Investigation and intervention recommended)

Rekomendasi:

1. Investigation: Analyze model features, check for proxy discrimination
2. Feature audit: Review fitur yang berkorelasi dengan Age (misal: Employment_Years)
3. Threshold adjustment: Pertimbangkan adjust approval threshold untuk fairness
4. Fairness constraints: Implement demographic parity atau equalized odds constraints
5. Monitoring: Deploy real-time bias monitoring
6. Periodic re-assessment: Re-audit tiap bulan untuk track improvement

5. PANDUAN VISUALISASI

5.1. Distribution Analysis Charts

1. Histogram (Overlapping):
 - Menampilkan distribusi credit score untuk Young (Navy Blue) dan Old (Grey)
 - Cara baca: Jika 2 histogram TERPISAH JAUH = bias kuat
 - Jika OVERLAP banyak = distribusi mirip, fair
2. Box Plot:
 - Menampilkan quartile (Q1, median, Q3) dan outliers
 - Cara baca: Bandingkan MEDIAN (garis tengah kotak)
 - Jika median Young >> median Old = bias
3. Violin Plot:
 - Kombinasi box plot + kernel density plot
 - Bagian LEBAR = banyak data di range tersebut
 - Cara baca: Lihat shape dan overlap
 - Jika shape TIDAK OVERLAP = bias kuat

5.2. Income vs Score Charts

1. Bubble Scatter Plot:
 - X-axis: Annual Income, Y-axis: Credit Score
 - Bubble size: Credit Score (makin besar = makin tinggi)
 - Warna: Navy (Young), Grey (Old)
 - Cara baca: Jika Grey SELALU DI BAWAH Navy = bias across ALL income levels
2. Age-Score Trend Line:
 - Line chart: Age (20-66) vs Average Score
 - Shaded area: Confidence band (standard deviation)
 - Vertical line: Age 45 (threshold)
 - Cara baca: Jika garis TERJUN di age 45 = clear bias pattern

5.3. Advanced Charts

1. Correlation Heatmap:
 - Matrix korelasi semua numerical features
 - Warna: Navy (high correlation), Grey (low)
 - Cara baca: Cari cell Age vs Credit Score
 - Nilai negatif tinggi (misal -0.9) = strong negative correlation = bias
 - Juga cari PROXY features (Employment_Years vs Age)
2. Fairness Radar Chart:
 - Pentagon dengan 5 fairness metrics
 - Reference line di 80 (threshold)
 - Cara baca: Jika blue line DI BAWAH threshold = FAIL metric tersebut
3. Sunburst Chart:
 - Hierarchical: Age Group -> Risk Category -> Loan Status
 - Interactive: Klik untuk drill-down

- Cara baca: Bandingkan proporsi warna per age group

6. REFERENSI DAN TINJAUAN PUSTAKA

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